C5.2 General requirements for snoop transactions

For each snoop transaction, the protocol specifies required and recommended behaviors.

Table C5-3 shows the required behavior for each snoop transaction:

Table C5-3 Required snoop transaction behavior

Snoop transaction	Must transfer data if dirty	End state must be Shared or Invalid	End state must be Invalid	End state must be Clean or Invalid
ReadOnce	Yes	-	-	-
ReadClean	Yes	Yes	-	-
ReadNotSharedDirty	Yes	Yes	-	-
ReadShared	Yes	Yes	-	-
ReadUnique	Yes	-	Yes	-
CleanInvalid	Yes	-	Yes	-
MakeInvalid	-	-	Yes	-
CleanShared	Yes	-	-	Yes

Note

If a cache line is in the Dirty state, and the associated cache does not assert the PassDirty snoop response, **CRRESP[2]**, the cache line can remain in the Dirty state. If a cache line is in the Dirty state and the associated cache does assert the PassDirty snoop response, then the cache line must move to a Clean or Invalid state.

The cache line end states in Table C5-3 are classified as follows:

Shared or Invalid

The snooped cache must broadcast a transaction before it can perform a store to the cache line. That is, the snooped cache must consider that another master can hold a copy of the cache line.

Invalid The snooped cache does not hold a copy of the line. This permits another agent to perform a store to the cache line.

Clean or Invalid

The snooped cache is not holding the cache line in a Dirty state. The snooped cache cannot perform a memory update, using a WriteBack or WriteClean transaction, until a later store to the cache line has occurred.

The following state changes must not occur due to a snoop transaction. A cache line must not move from:

- the Invalid state to any Valid state
- a Clean state to a Dirty state
- a Shared state to a Unique state
- the UniqueDirty state to the UniqueClean state.

Note

A cache line must not move from the UniqueDirty state to the UniqueClean state because such a transition would indicate that the interconnect has taken responsibility for writing back the cache line to main memory. Therefore, the cached master must not issue a WriteBack or WriteClean transaction without requesting permission to store to the cache line using an appropriate transaction.

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The cache line end state that is permitted as a result of a snoop transaction is dependant on:

- the state of the cache line before the snoop
- the snoop transaction that is issued.

Table C5-4 shows the permitted end states for valid combinations of the initial state and the issued snoop transaction. Combinations of initial state and end state that are not permitted as a result of a snoop transaction are excluded from Table C5-4.

A WriteBack, WriteClean or Evict transaction can occur while a snoop transaction is in progress. Table C5-4 does not include the state transition that can occur as a result of these write or evict transactions occurring. To understand such a scenario, the state transition for the WriteBack, WriteClean or Evict transaction must be applied, followed by the snoop transaction state transition that Table C5-4 shows.

The following abbreviations are used for the cache line states:

UC UniqueClean.
UD UniqueDirty.
SC SharedClean.
SD SharedDirty.
I Invalid.

Table C5-4 Permitted end states for combinations of initial state and snoop transaction

Cache line state		Permitted for snoop transaction			
Initial	End	ReadOnce	ReadClean ReadNotSharedDirty ReadShared	ReadUnique CleanInvalid MakeInvalid	Clean Shared
I	I	Yes	Yes	Yes	Yes
UC	I	Yes	Yes	Yes	Yes
	UC	Yes	-	-	Yes
	SC	Yes	Yes	-	Yes
UD	I	Yes	Yes	Yes	Yes
	UD	Yes	-	-	-
	SC	Yes	Yes	-	Yes
	SD	Yes	Yes	-	-
SC	I	Yes	Yes	Yes	Yes
	SC	Yes	Yes	-	Yes
SD	I	Yes	Yes	Yes	Yes
	SC	Yes	Yes	-	Yes
	SD	Yes	Yes	-	-

The requirements for the IsShared and PassDirty snoop response bits are as follows:

- if the end state of the cache line is any Valid state, the IsShared snoop response bit must be asserted
- if the cache line moves from a Dirty state to a Clean state, the PassDirty snoop response bit must be asserted
- if the line moves from a Dirty state to the Invalid state, as a result of any snoop transaction except MakeInvalid, the PassDirty snoop response bit must be asserted

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- if the cache line moves from a Dirty state to the Invalid state, as a result of a MakeInvalid snoop transaction, the PassDirty snoop response bit can be asserted or deasserted.

The permitted state changes in Table C5-4 on page C5-214 are combined with these requirements in Table C5-5 to show the permitted state changes and associated snoop response bits.

Table C5-5 Associated snoop responses for combinations of initial state and snoop transaction

Cache line state		Permitted for snoop transaction				Snoop Response	
Initial	**End**	**ReadOnce**	**ReadClean ReadNotSharedDirty ReadShared**	**ReadUnique CleanInvalid MakeInvalid**	**CleanShared**	**PassDirty**	**IsShared**
I	I	Yes	Yes	Yes	Yes	0	0
UC	I	Yes	Yes	Yes	Yes	0	0
	UC	Yes	-	-	Yes	0	1
	SC	Yes	Yes	-	Yes	0	1
UD	I	Yes	Yes	Yes	Yes	1[a]	0
	UD	Yes	-	-	-	0	1
	SC	Yes	Yes	-	Yes	1	1
	SD	Yes	Yes	-	-	0	1
SC	I	Yes	Yes	Yes	Yes	0	0
	SC	Yes	Yes	-	Yes	0	1
SD	I	Yes	Yes	Yes	Yes	1[a]	0
	SC	Yes	Yes	-	Yes	1	1
	SD	Yes	Yes	-	-	0	1

a. For a MakeInvalid snoop transaction these PassDirty responses are also permitted to be 0.

C5.2.1 Channel activity

The required channel activity is the same for all snoop transactions:

- the address is received on the AC channel
- the response is returned on the CR channel
- the data is provided, if required, on the CD channel.

The DataTransfer snoop response bit **CRRESP[0]**, indicates if a data transfer is required.

The snoop response on CR and the snoop data on CD must only be provided after the **ACVALID**/**ACREADY** handshake occurs.

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C5.2.2 Snoop data transfers

A cached master can provide the data value of a cache line. The DataTransfer snoop response bit indicates that the data value of the cache line is to be transferred.

If a cached master receives a snoop transaction other than MakeInvalid, for a cache line that is in a Dirty state, then the cached master must ensure that the data value is available so that the original transaction can complete. The cached master can ensure that the data value is available by:

- returning the data when it completes the snoop transaction
- carrying out a memory update, using a WriteBack or WriteClean transaction, before responding to the snoop transaction.

Note

When a cached master holds a cache line in a Dirty state, the cache line might be the only up to date copy of that address location. Therefore, the data must be made available to any snoop transaction other than a MakeInvalid snoop transaction.

Typically, data is transferred for the following read snoop transactions:

- ReadOnce
- ReadClean
- ReadNotSharedDirty
- ReadShared
- ReadUnique.

Table C5-6 shows protocol-recommended data transfer behavior for snoop transactions, based on the assumption that better system performance and lower power operation is achieved by providing data in response to these snoop transactions. This behavior is not mandatory, and alternative schemes can be implemented.

Table C5-6 Recommended data transfer behavior for snoop transactions

Snoop transaction	Data transfer if cache line is Clean	Data transfer if cache line is Dirty
ReadOnce	Yes	Yes
ReadClean	Yes	Yes
ReadNotSharedDirty	Yes	Yes
ReadShared	Yes	Yes
ReadUnique	Yes	Yes
CleanInvalid	No	Yes
MakeInvalid	No	No
CleanShared	No	Yes

C5.2.3 Memory update in progress

The protocol ensures that two components cannot update the same area of main memory at the same time.

If a snooped master receives a snoop transaction when it is updating main memory using either a WriteBack or WriteClean transaction, then it is the responsibility of the snooped master to ensure that no other master can update the same area of main memory at the same time. The snooped master achieves this by one of the following:

- giving a snoop response with PassDirty deasserted and IsShared asserted, which does not pass permission to store to the line and does not pass responsibility for updating memory
- delaying the snoop response until the snooped master has completed the update to main memory.

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When a snooped master is passing the permission to store to a cache line, by sending a suitable snoop response, all write transactions to update main memory must have completed before the cycle in which the snoop response is given on the CR channel.

C5.2.4 WasUnique snoop response

The WasUnique snoop response, **CRRESP[4]** indicates that the snooped cache line was held in a Unique state before the snoop transaction.

The WasUnique snoop response must only be asserted if the cache line was held in a Unique state and therefore, no other cache can have a copy of the cache line. A WasUnique response indicates that the interconnect does not have to carry out further snoop transactions to other cached masters because no other cache can hold a copy of the data.

A cached master does not have to generate the WasUnique response. The protocol permits **CRRESP[4]** to be fixed as deasserted. However, always deasserting WasUnique in this way can result in the cache line being provided to the initiating master as Shared, when it could have been provided as Unique. This might result in additional caches being snooped unnecessarily.

C5.2.5 Non-blocking requirements for a snooped master

To ensure transactions always progress through a system, the protocol defines rules for snooped masters and the interconnect. The rules stipulate which transactions must always make progress and which transactions can wait for others to complete.

The rules for the interconnect are defined in Chapter C6 *Interconnect Requirements*. See *Non-blocking requirements* on page C6-237.

The rules that apply to a cached master are:

- A master must complete any snoop transaction, to any address, before any of the following transactions, issued by the master, can be guaranteed to complete:
  - Any transaction, to any address, issued on the AR channel
  - A WriteUnique or WriteLineUnique transaction, to any address, issued on the AW channel.

    Note

    See also *Restrictions on WriteUnique and WriteLineUnique Usage* on page C4-205.

- A master is permitted to wait for the following transactions to complete, to any address, before completing a snoop transaction:
  - WriteNoSnoop
  - WriteBack
  - WriteClean
  - Evict.
- If the response to a snoop transaction could result in the interconnect generating a write to main memory, or another master being given permission to write the cache line, then the master being snooped must complete any WriteBack or WriteClean transaction that is in progress for the cache line before it provides a response to the snoop transaction.
- A master must not wait for a WriteUnique or WriteLineUnique transaction to complete before completing a snoop transaction. If a snoop transaction is received by a master and a WriteUnique or WriteLineUnique transaction is in progress then the snoop transaction must be completed without the use of the AW and W channels.

  Note

  This requirement means that if a master has a WriteUnique or WriteLineUnique transaction in progress for any cache line that is in a Dirty state, and it receives a snoop transaction other than a MakeInvalid transaction, then it must return the data on the CD channel.

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The diagram in Figure C5-1 shows the non-blocking requirements.

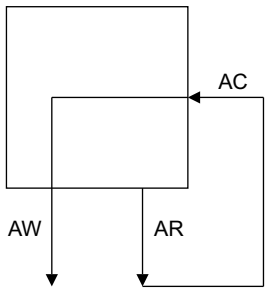


Figure C5-1 Required transaction channel ordering

In summary, the requirements are:

- any transaction on the AR channel can be stalled waiting for a transaction on the AC channel
- any snoop transaction on the AC channel can be stalled waiting for a write transaction on the AW channel, except for a WriteUnique or WriteLineUnique transaction.

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